

Trans-SAM: Transfer Segment Anything Model to medical image segmentation with Parameter-Efficient Fine-Tuning

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ABSTRACT

Recently, the Segment Anything Model (SAM) has gained substantial attention in image segmentation due to its remarkable performance. It has demonstrated impressive zero-shot capabilities through interactive segmentation. However, due to the predominance of natural images in the training data, it often exhibits unsatisfactory performance when directly applied to medical image segmentation tasks. Training SAM from scratch in the medical domain is computationally resource-intensive and data-demanding. Additionally, when fine-tuning pre-trained SAM, the limited quantity of medical data will potentially lead to catastrophic forgetting. To address these issues, we proposed Trans-SAM, which utilizes Parameter-Efficient Fine-Tuning (PEFT) to transfer SAM into the medical image segmentation. The key novelties include the Intuitive Perceptual Fine-tuning (IPF) adapter, which directly integrates input image features into each layer of the encoder, and the Multi-scale Domain Transfer (MDT) adapter, a convolution-based mechanism designed to infuse inductive biases into the SAM. With our method, the general features extracted by pre-trained SAM can be better transferred to the medical domain. Moreover, our method can achieve satisfactory performance with a minimal amount of data during the fine-tuning. We conducted extensive evaluations on six medical datasets including different organs and modalities. The experimental results demonstrate that Trans-SAM shows superior performance compared to state-of-the-art PEFT methods. It significantly improves the performance of SAM for medical image segmentation. The code will be available at <https://github.com/wuyanlin-wyl/Trans-SAM>.

1. Introduction

Medical image segmentation is of great significance in clinical practice, which can provide auxiliary diagnostic information for doctors [1–5]. In recent years, deep learning has been widely used in medical image segmentation due to its powerful feature learning ability [6–9]. Currently, state-of-the-art medical segmentation methods can be broadly classified into two main categories: Convolutional Neural Networks (CNNs) [10–14] and Transformer-based architectures [15–17]. However, most of them depend on careful design of the model structure, significantly hindering model versatility and scalability.

The recent success of foundation models, SAM [18], SegGPT [19], and SEEM [20] have been developed, providing a unified framework for various visual tasks. They exhibit excellent generalization capabilities and perform robustly in zero-shot, one-shot, and few-shot scenarios, reducing the burden of network engineering and model deployment. However, these foundation models cannot be directly applied to medical tasks for the following reasons: (1) Visual foundation models are mostly trained on natural images, while medical images have low contrast and diverse anatomical structures. Applying them directly to

medical images would greatly impact their performance. (2) The majority of existing medical image segmentation datasets were constructed without corresponding prompt points or boxes, making it difficult to use interactive model for fine-tuning. (3) The segmentation results of these models lack semantic information, and the same object in different images may be assigned with different semantics.

To address the above challenges, we proposed Trans-SAM to adapt SAM to medical image segmentation. Our methods can automatically segment the lesions or organs and assign a specific semantic label to these segmentation results. As shown in Fig. 1, Trans-SAM achieves, and even surpasses, the performance of other networks that undergo full fine-tuning by training only a small portion of parameters, demonstrating the efficiency and effectiveness of our approach. Compared with previous approaches, the contributions can be summarized as follows:

1. We proposed a novel parameter-efficient fine-tuning approach named Intuitive Perceptual Fine-tuning (IPF) adapter, which introduces a small number of parameters to fine-tune the SAM for medical image segmentation tasks. It embeds the initial patch embedding into adapter

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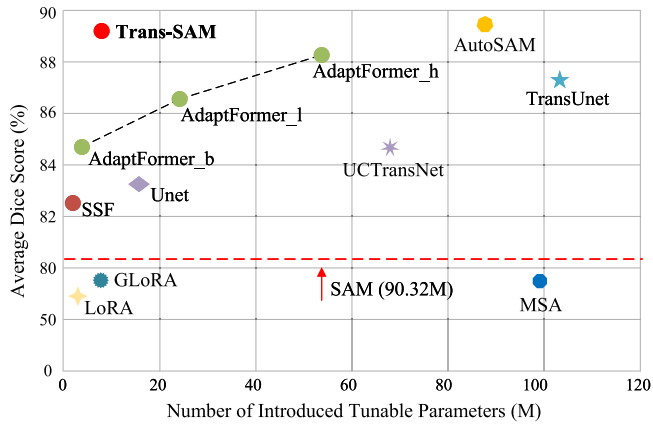


Fig. 1. Comparison of Trans-SAM and existing methods on the Kvasir datasets. Our approach attains a trade-off between trainable parameters and performance compared to other parameter-efficient fine tuning methods.

of each encoder layer, allowing even the adapters in deep layers to directly perceive the original features from medical images, rather than fine-tuning the features that have been extracted through multiple layers.

2. Further, we presented the Multi-scale Domain Transfer (MDT) adapter, a novel architecture comprised of convolutional layers. This design substantially incorporates inductive bias into the SAM, thereby enhancing the model's efficiency and effectiveness in conducting fine-grained analysis and processing. Additionally, three parallel pathways with different receptive fields allows the adapter to understand and integrate information across various scales more effectively.

3. Extensive experiments and analyses on various medical image segmentation tasks have shown that the proposed method surpasses other state-of-the-art models in performance. Moreover, experimental results demonstrate that Trans-SAM effectively reduces the SAM's dependency on labeled data throughout the training stage. More importantly, our method achieves state-of-the-art performance in the semantic segmentation without prompts.

2. Related work

2.1. Medical image segmentation

Medical image segmentation is crucial in computer-aided diagnosis. Since the introduction of UNet [21], this field has seen greatly developed. Improved versions of UNet such as UNet++ [10] and UNet3+ [11] enhance the information transmission capability of skip connection. MultiResUnet [22] utilize a multi-resolution approach to extract multi-scale information. Att-UNet [13] applies attention gates to weight feature maps at the same level. With the successful application of ViT [23], researchers have begun to explore the integration of Transformers into medical image segmentation [24–26]. For example, TransUNet [15] incorporates Transformer after convolutional layers to extract global information from lower-resolution feature maps. SwinUNet [27] employs Swin Transformers [28] to construct an all-Transformer U-shaped architecture. TransFuse [16] simultaneously builds CNN and Transformer encoders to fully exploit their advantages, reducing network depth while capturing both high-level and low-level information. UCTransNet [17] proposes a multi-scale channel cross-fusion module and redesigns the skip connection, leveraging Transformers to extract relationship from different layers. Despite the success of these methods in some tasks of medical image semantic segmentation, they often face challenges such as data scarcity and model overfitting. In recent years, prototype-based semantic segmentation methods have gained attention. For instance, Zhou et al. proposed

a non-parametric semantic segmentation method [29], which uses fixed, non-learnable prototypes to represent each class, instead of using learned class weights or query vectors. By calculating the similarity between image pixels and class prototypes, classification is performed, thus avoiding the complexity of class representation in traditional methods. Similar to the core idea of the prototype-view, we effectively integrate raw features of medical images into the segmentation model through the IPF adapter, significantly enhancing the model's performance and improving segmentation results in the case of limited data.

2.2. Segment anything model

SAM [18] is the first vision foundational model and it was applied to a variety of downstream tasks, such as image restoration [30], detection [31], dataset construction [32], and robotic surgery [33]. In addition, Cheng et al. proposed the SAM-Track framework, which combines the SAM and the DeAOT tracking model to achieve precise segmentation and tracking of objects in video [34]. Sun et al. proposed the Explain Any Concept method, which leverages the SAM to achieve concept-driven explainable AI [35]. Wang et al. proposed SAM-CLIP, a unified model that efficiently merges the capabilities of SAM and CLIP through multi-task learning, continual learning, and distillation techniques, extending SAM to the multimodal domain [36]. However, the performance of SAM in medical image segmentation is unsatisfactory. For instance, Huang et al. [37] demonstrated that SAM cannot achieve good segmentation accuracy in specific medical modalities. Shi et al. [38] found that the zero-shot performance of SAM on medical images remains constrained, especially in tasks involving continuous segmentation structures like retinal vessels. And Cheng et al. [39] found its zero-shot prediction accuracy was lower than commonly segmentation models. Inspired by the fine-tuning of Large Language Models [40], some researchers have explored fine-tuning SAM to enhance its applicability in medical image segmentation. For instance, Ma et al. [41] proposed MedSAM, which freezes the image encoder and prompt encoder, fine-tunes the decoder and adopts a box-based prompt method. Similarly, Li et al. [42] also fine-tuned SAM with a box-based prompt method, achieving good performance on five datasets. In addition, Wu et al. [43] proposed Medical sam adapter, which showed superior performance in 19 medical image segmentation tasks.

Although significant progress has been made in enhancing the performance of SAM, there are still some challenges and limitations. One is the lack of explicit semantic for different segmentation results. Additionally, methods based on prompts are challenging to directly apply to the majority of medical image segmentation tasks. Most importantly, they often struggle to generalize effectively across different datasets and domains. Therefore, it is necessary to further study the application of SAM in medical image segmentation.

2.3. Parameter efficient fine tuning

With the increase of data and model scale, the paradigm of pre-training large models and transferring them to downstream tasks is gaining popularity. During the transfer learning, researchers proposed parameter-efficient fine-tuning (PEFT), which originally emerged in the field of natural language processing [44,45]. Specifically, PEFT transfer the general features learned by pre-trained large models to task-specific features. For example, Visual Prompt Tuning (VPT) [46] freezes the weights of the pre-trained model, and only introduces a small number of trainable parameters as prompts, achieving excellent performance and storage efficiency. Similarly, Pro-tuning [47] constructs multiple stages of prompt blocks and inserts them into existing frozen pre-trained visual models to generate prompt information. LoRA [40] proposes Low-Rank Adaptation, which keeps the pre-trained model

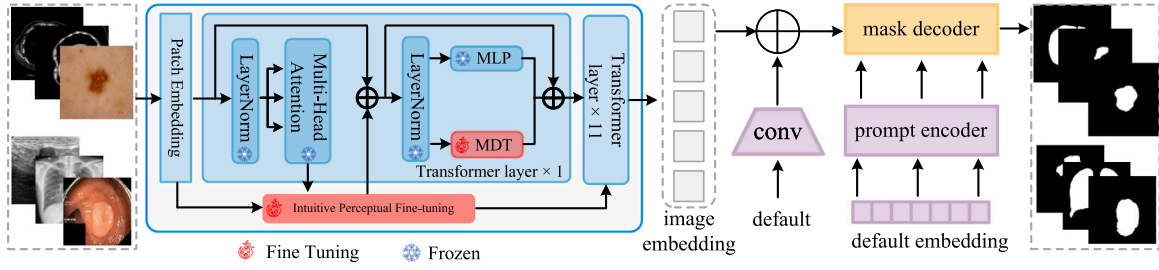


Fig. 2. The overall structure pipeline of our method. We frozen the image encoder of SAM and fine-tuned the model with IPF and MDT adapter. Additionally, the input of the prompt encoder is set to default, achieving automated segmentation without prompts.

weights frozen and introduces externally trainable low-rank decomposition matrices. Adapter [48] utilize bottleneck modules to effectively adapt pre-trained Vision Transformers to various visual tasks. On the other hand, SSF [49] applies scale and shift operations to the extracted features from the pre-trained model, modulating them into a specific feature space that is better suited for downstream tasks.

In medical image segmentation, MSA [43] incorporates Adapter into the pre-trained SAM, successfully handling segmentation tasks for various modalities of medical images. MedTuning [50] introduces a novel framework, achieving intra-stage feature enhancement and inter-stage feature interaction. It achieves better segmentation results in 3D medical image segmentation tasks than previous state-of-the-art parameter-efficient transfer learning methods. SAMed [51] aims to explore a novel research paradigm focused on customizing large-scale models specifically for medical image segmentation. However, the aforementioned PEFT approach predominantly employs fully connected operations, and the effectiveness of convolutional in feature adaptation has not been thoroughly investigated [52]. Additionally, these methods lack effective feature selection and enhancement during feature adaptation. This limitation constrains their performance when dealing with challenging tasks, thereby reducing the model's overall efficacy. To address these issue, we introduce convolution operation and attention mechanisms to promote model more accurately modeling medical domain knowledge. With this approach, SAM can be more adeptly transferred to medical image segmentation tasks.

3. Method

In this section, we will first briefly introduce the overall structure of our method. Following that, we will provide a comprehensive introduction to the proposed PEFT. The pipeline of our structure is illustrated in Fig. 2.

3.1. Overall structure

As shown in Fig. 2, our method consists of three main components. The first component is an image encoder used for extracting image features, which is a pre-trained ViT architecture based on MAE [53]. We freeze the parameters of image encoder and introduce IPF and MDT adapter to adapt the features. The second part is the prompt encoder, the original SAM includes two types of prompts: sparse prompts (point, box, and text) and dense prompts (mask). The point and box in sparse prompts are mapped to 256-dimensional vectors in the form of position encoding, while the text context leverages the text encoder in the pre-trained CLIP [54] model for representation. The dense prompt is also mapped to a 256-dimensional vector through convolution. To enable most existing medical image segmentation datasets to be directly applied for fine-tuning with Trans-SAM, and to facilitate further research extensions, we did not remove the prompt encoder. Instead, we set the input of the prompt encoder to be empty, allowing the model to automatically fill the prompt embeddings. The last part is mask decoder. Each decoder layer performs four steps: (1) self-attention on the tokens, (2) cross-attention from the tokens (as queries) to the image

embedding, (3) multi-layer perceptron (MLP) updates for each token, and (4) cross-attention from the image embedding (as queries) to the tokens. In the cross-attention, the image embedding is treated as a set of $64 \times 64 \times 256$ -dimensional vectors. Every self-attention, cross-attention and MLP has a residual connection, layer normalization, and dropout. The next decoder layer obtains updated tokens and updated image embedding from the previous layer. Then, upsampling is performed via transpose convolution, and the final output mask is obtained by the segmentation head.

3.2. Intuitive perceptual fine-tuning adapter

To allow the deep encoder to directly perceive the initial features of medical images, rather than relying on information extracted after multiple layers, we designed the IPF adapter. This adapter enhances the model's effective use of the initial information in medical images while reducing the number of introduced parameters. The specific structure of the Intuitive Perceptual Fine-tuning Adapter is shown in Fig. 3. Existing methods typically add an adapter [48] to every layer of the encoder during model fine-tuning, thereby adjusting the features obtained at each layer. However, as the number of layers increases, there is an inevitable loss of information from the target domain. Taking this insight into account, we aimed to allow every pre-trained layer in the encoder to intuitively perceive the image content from the target domain during the fine-tuning process. To achieve this, we introduced an additional branch path into the adapter.

Specifically, for an image encoder with N Transformer layers, it divides the input image $I \in \mathbb{R}^{3 \times H \times W}$ into M patches $X_0 \in \mathbb{R}^{M \times d}$ through the patch embedding layer. Then the original SAM image encoder processes X_0 layer by layer.

$$X_l = \text{MHSA}_l(\text{LN}_l(X_{l-1})), \quad l = 1, 2, \dots, N \quad (1)$$

where LN stands for layer normalization, and MHSA represents multi-head self-attention. In MHSA, tokens will be projected to obtain Q (Query), K (Key), and V (Value). The specific operations can be described as follows:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

d_k represents the dimension of the K and acts as a scaling factor to maintain the stability of gradients.

In the process of enhancing the Transformer encoder, besides individually adding an adapter to each Transformer layer, we duplicate the patch embeddings X_0 obtained before the image enters the Transformer layers. One copy is fed into the MHSA within the Transformer layer for computation, while the other is processed through a shared projection layer to obtain intuitive features X_{int} , which are then stored. As shown in Fig. 3, in every subsequent layer of the encoder, this part of the image embedding information from the initial phase of the network is integrated into the adapter, merging with the feature maps fine-tuned at each layer to better assist the fine-tuning process of each layer's adapter. Furthermore, after feature fusion, we introduce the triplet

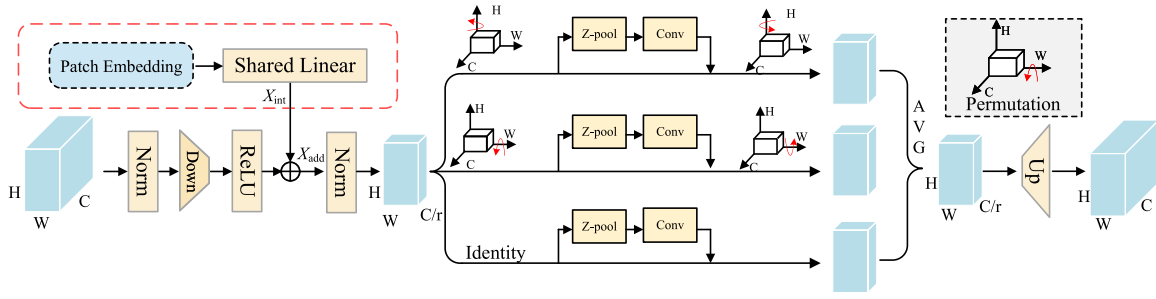


Fig. 3. Overall structure of the Intuitive Perceptual Fine-tuning (IPF) Adapter, integrating an additional branch path to enhance feature representation. It begins with patch embedding and progresses through a shared linear projection. Initial image embeddings are merged at each pre-trained layer.

attention mechanism [55], leveraging spatial and channel attention, to enable features merged from the current and initial stages to be transferred to the target domain more rapidly and accurately:

$$X' = \text{TA}(W_1 X + X_{int}) W_2 \quad (3)$$

where $W_1 \in \mathbb{R}^{d \times \frac{d}{r}}$ and $W_2 \in \mathbb{R}^{\frac{d}{r} \times d}$ are responsible for downsampling and upsampling the feature maps, respectively. X_{int} is output by a shared linear layer, with different Transformer layers sharing an initial projection layer, enabling the introduction of initial medical image features at various levels of the model. This improves the model's ability to learn and understand the fine-grained features of medical images at different depths. Triplet attention consists of three parallel branches. For the fused feature $X_{add} \in \mathbb{R}^{C \times H \times W}$, the first two paths perform two transpose operations to obtain $X_1 \in \mathbb{R}^{W \times H \times C}$ and $X_2 \in \mathbb{R}^{H \times W \times C}$ respectively, followed by Z-Pool to get $X'_1 \in \mathbb{R}^{2 \times H \times C}$ and $X'_2 \in \mathbb{R}^{2 \times W \times C}$. Here, the Z-Pool layer reduces the tensor from the first dimension, preserving the rich representation of feature maps while reducing its depth to decrease the model's computational complexity:

$$\text{Z-pool}(x) = [\text{MaxPool}_{0d}(x), \text{AvgPool}_{0d}(x)] \quad (4)$$

Subsequently, attention weights are generated by a convolutional layer with a 7×7 kernel size and a sigmoid function, and are applied separately to X_1 and X_2 . These two paths are responsible for capturing cross-dimensional interactions between channel C and spatial dimensions. The last branch, similar to CBAM [56], is used to construct Spatial Attention. The outputs of the three branches are aggregated using an averaging operation. By capturing the interplay between the spatial and channel dimensions of the input feature maps, this approach addresses the issue of lack of interdependency between space and channel in traditional attention mechanisms.

3.3. Multi-scale domain transfer adapter

To enable SAM to capture lesion features of varying sizes and shapes through a multi-scale processing mechanism, and to combine the local inductive bias properties of convolutions to enhance SAM's local detail perception and overall accuracy in medical image analysis, we designed the MDT adapter. This adapter improves SAM's accuracy and applicability in the medical image domain. As shown in Fig. 2, this structure is added in parallel to the original Transformer's MLP module, jointly processing the previously obtained X' . The MDT Adapter is characterized by being entirely composed of convolutions, unlike the original adapter which is composed of fully connected layers. Convolutional operations, by capturing local features and hierarchical learning, can effectively transfer the feature representation learned from natural images to medical domain, thereby improving the model's generalization ability. Furthermore, important features in medical images may exist at different scales, such as small-scale features at the pixel level and large-scale features at the organ level. As illustrated in Fig. 4, we introduce three paths with convolutional kernels of different

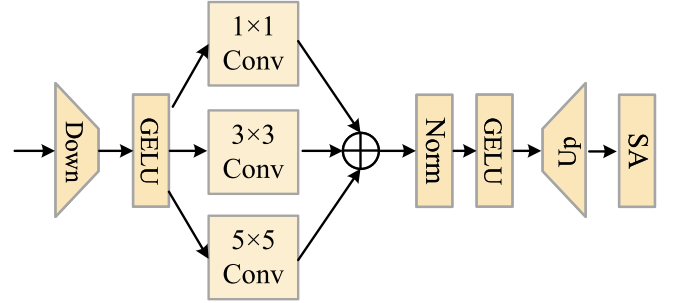


Fig. 4. Architecture of the Multi-scale Domain Transfer (MDT) Adapter, featuring parallel convolutional paths with 1×1 , 3×3 and 5×5 kernels to capture multi-scale features, integrated with spatial attention for enhanced medical image segmentation.

sizes to accommodate the recognition of these different scale features, thereby enhancing the model's performance in complex medical image segmentation tasks.

Specifically, for an input feature map $X' \in \mathbb{R}^{W \times H \times C}$, dimension reduction is first performed by a convolution with a kernel size of 1×1 , followed by a GELU activation and then it is fed into a multi-scale processing path. The multi-scale processing path includes three branches with kernel sizes of 1×1 , 3×3 and 5×5 , respectively, which are used to capture important features at different scales during the fine-tuning process. After merging the feature maps from these three paths, the combined feature map undergoes normalization and a GELU activation, then it is fed into a convolution with a kernel size of 1×1 to restore the channel dimension. Finally, the transfer features are further enhanced through a spatial attention mechanism. The spatial attention mechanism adopts the calculation of spatial attention from CBAM [56], and the calculation process can be expressed as:

$$\text{SA}(\tilde{X}) = \sigma \left(f^{7 \times 7} \left(\left[\text{AvgPool}(\tilde{X}); \text{MaxPool}(\tilde{X}) \right] \right) \right) \quad (5)$$

where σ represents the activation operation, and $f^{7 \times 7}$ represents a convolution operation with a kernel size of 7×7 , which is used to compress the feature map after global average pooling and global max pooling along the channel dimension.

4. Experiment

In this section, we first introduced the publicly available medical image segmentation datasets in our study. Then, we provided details of the experimental setup. Furthermore, to validate the effectiveness and superiority of our proposed method, we conducted comprehensive experiments to analyze its performance.

4.1. Datasets

LiTS [57]: The Liver Tumor Segmentation Challenge (LiTS) is a publicly available medical image segmentation dataset primarily used

for automatic liver and tumor segmentation. The dataset consists of 131 3D CT convolutional images along with corresponding labels, covering the liver and lesion regions of different patients. This dataset has become a recognized benchmark widely applied in various liver automatic segmentation studies.

ISIC [58]: This dataset contains 2594 skin lesion images captured from real patients using a dermatoscope equipped with a digital camera. Each image has been annotated by a professional physician to mark the area of the skin lesion, and all data have been reviewed and managed by professional dermatologists with knowledge of dermatoscopy.

Kvasir [59]: Kvasir-SEG is an open-source dataset manually annotated and verified by an experienced gastroenterologist. The dataset contains 1000 images of polyps and their corresponding masks, with image resolutions ranging from 332×487 to 1920×1072 . The aim of creating this dataset is to promote the development and progress of polyp detection tasks.

BUSI [60]: The BUSI dataset collects ultrasound images of breasts from women aged between 25 and 75 years old. The dataset includes 780 images from 600 female patients, which are divided into three categories: normal, benign, and malignant. Among them, there are 133 normal cases, 437 benign tumors, and 210 malignant tumors. The average image size is 500×500 pixels.

CXML [61,62]: Chest Xray Masks and Labels contains 138 posterior-anterior X-rays, of which 80 X-rays are normal and 58 X-rays are abnormal with manifestations of tuberculosis. All images are de-identified and available in DICOM format.

FML [63]: Finding and Measuring Lungs in CT Data is a Kaggle competition dataset with data from four patients. The dataset consists of a collection of 2D and 3D images, including segmentation masks corresponding to lung parenchyma.

4.2. Implementation details

In the experiment, we split every dataset into train, validation, and test set in an 8:1:1 ratio. All image resolutions are uniformly resized to 256×256 . Additionally, data augmentation techniques such as Gaussian blurring, random flipping, and affine transformations were applied during the training stage. The pre-trained SAM with the ViT-Base was chosen as the foundational model. For hyperparameter tuning, we set the learning rate to 0.0001, and the maximum training epochs was set to 200. The learning rate decay are linear warming up and cosine annealing, with a warming up epoch of 10. The batch size was set to 8. All models were optimized using the AdamW [64] optimizer, and the weight decay was set to $1e-8$. To enhance training stability and improve the segmentation performance of the model, we adopted a joint loss function combining dice loss and binary cross-entropy loss. All experiments were conducted on a GeForce RTX 3090 with 24 GB memory. For the model evaluation, we adopt the Dice coefficient (DICE), Intersection over Union (IOU) and Accuracy(ACC) as the evaluation metrics.

4.3. Comparisons with other fine-tuning strategies

We also compared our proposed method with other fine-tuning strategies. Specifically, we integrated various visual fine-tuning strategies into the image encoder of SAM, evaluating the effectiveness of these strategies in adapting SAM for medical image segmentation tasks. The experimental results are shown in Table 1. Firstly, we performed full fine-tuning on SAM and used this result as a baseline reference. From Table 1, the LoRA [40] and its improved version GLoRA [65] achieved unsatisfactory performance compared to other fine-tuning strategies. In Kvasir and CXML, their performance was even lower than full fine-tuning. For VPT [46], we adopted two versions: VPT_shallow, which introduces learnable tokens only in the first-layer Transformer, and VPT_deep, which adds learnable tokens in each layer. However, the

Table 1

Comparisons with Other Fine-tuning Strategies. All the metrics are based on the DICE performance. VPT_s and VPT_d represent the version of shallow and deep, respectively. And the Adapt_b, Adapt_l, Adap_h represent the parameter version of base, large and huge.

Methods	LiTS	ISIC	Kvasir	BUSI	CXML	FML
Full_finetuning	96.54	88.06	80.25	56.83	94.15	93.51
LoRA [40]	94.87	89.31	69.85	64.22	93.18	95.51
GLoRA [65]	97.11	88.37	70.88	92.52	93.29	93.95
VPT_s [46]	92.53	89.08	59.91	57.08	88.78	93.27
VPT_d [46]	93.27	89.23	69.08	57.67	89.13	93.65
SSF [49]	94.05	90.38	82.64	56.64	94.39	95.33
Adapt_b [48]	96.37	90.91	84.55	68.23	95.28	95.55
Adapt_l [48]	96.58	91.35	86.27	72.11	95.21	96.49
Adapt_h [48]	96.85	91.66	88.19	71.53	95.38	96.27
Trans-SAM	97.09	91.89	89.09	73.01	94.99	96.58

performance of VPT was unsatisfactory on most tasks, and it even exhibited significant performance degradation in some tasks, performing worse than the full fine-tuning strategy. Although the SSF [49] achieved performance improvement on the ISIC, Kvasir, and FML datasets, the improvements were limited and the strategy showed poor robustness, with performance degradation on the other three datasets.

For the AdaptFormer [48], we conducted experiments on three versions: the base parameter version (Adapt_b, 0.47 M), the large parameter version (Adapt_l, 28.36 M), and the huge parameter version (Adapt_h, 57.02 M). In the experiments, Adapt_b achieved performance improvements on all six datasets. With increased model sizes, performance further improved on most datasets. However, there was a slight performance decline on the LiTS and CXML datasets as the number of parameters increased. This indicates that AdaptFormer does indeed have a certain level of effectiveness, but the stability of its performance improvement decreases as the number of parameters increases. In comparison, Trans-SAM achieved more noticeable performance improvements than other methods on six datasets.

4.4. Comparisons with other state-of-the-art methods

Quantitative Comparison: We conducted a comprehensive evaluation of our method against several state-of-the-art models across multiple datasets. The results, presented in Table 2, demonstrate that our proposed method exhibits significant competitiveness in medical image segmentation. Our model outperformed state-of-the-art counterparts on the majority of datasets across the six medical image segmentation tasks. These experimental results provide strong evidence that our method has effectively improved the capabilities of SAM in medical image segmentation. Furthermore, our method achieves the objective of employing a single model for segmenting multiple medical images, while consistently delivering excellent performance.

Additionally, the parameters of Trans-SAM is 96.96 M, slightly higher than SAMed's 90.47 M and MedSAM's 93.74 M. However, when performing inference on a single image, Trans-SAM takes 0.306 s, which is faster than SAMed's 0.409 s and MedSAM's 0.508 s. Although MedSAM performs better than Trans-SAM on certain datasets, it has issues such as not maintaining consistent semantics, the inability to batch process, and difficulties in selecting prompts and bounding boxes when the contrast between the target and surrounding tissues is extremely low. Therefore, Trans-SAM still holds significant comparative advantages.

Qualitative Comparison: We also conducted a qualitative comparison between our method and the current state-of-the-art models. The comprehensive comparison results are shown in Fig. 5, demonstrating that our proposed method performs effectively in segmenting the target region, even in challenging scenarios, such as low contrast between the target tissue and surrounding structures. This capability holds particular significance in the medical imaging, where subtle

Table 2

Quantitative comparison between SOTA methods and Trans-SAM on Medical Image Datasets. All the metrics are based on the DICE performance.

Methods	#P(M)	Inference(s)	LITS	ISIC	Kvasir	BUSI	CXML	FML
UNet [21]	17.27	0.213	95.54	87.77	83.92	61.09	94.56	88.74
Att-Unet [13]	34.88	0.256	95.71	88.03	86.69	65.14	94.50	89.42
TransUNet [15]	105.3	0.208	96.38	90.55	86.91	75.21	94.85	91.54
TransDeepLab [66]	17.49	0.251	95.88	89.22	71.63	62.29	94.35	93.96
HiFormer [67]	23.25	0.314	96.03	89.16	83.21	66.11	94.21	77.36
UCTransNet [17]	66.49	0.255	96.12	90.63	84.91	69.51	94.47	93.28
TransFuse [16]	26.28	0.402	96.25	90.21	82.78	70.54	94.28	92.47
AutoSAM [68]	90.82	0.311	94.77	91.74	89.61	73.55	94.59	78.74
MSA [69]	100.9	0.291	96.75	87.93	70.36	56.19	93.06	93.26
SAMed [70]	90.47	0.409	96.39	90.75	88.28	71.33	94.89	95.71
MedSAM [71]	93.74	0.508	96.87	91.35	90.13	75.53	94.67	96.31
Trans-SAM (Ours)	96.96	0.306	97.09	91.89	89.09	73.01	94.99	96.58

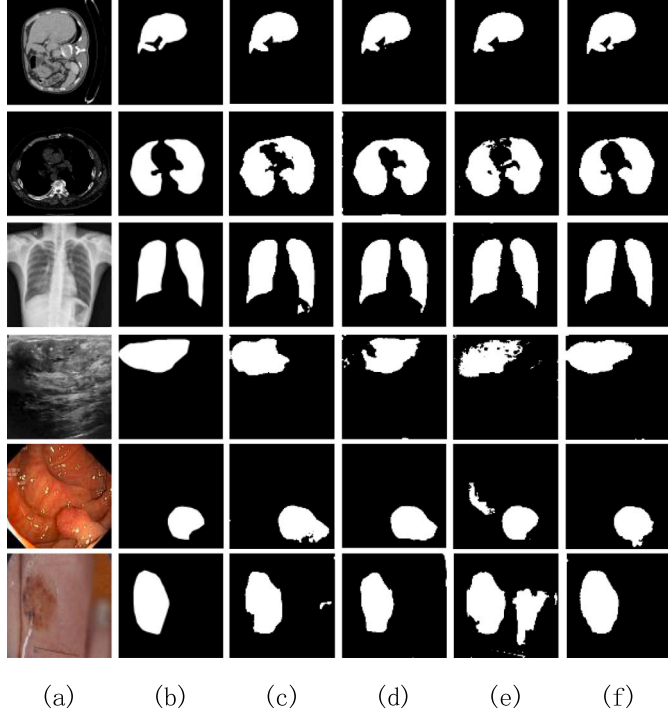


Fig. 5. Comparison of segmentation results on the CXML, BUSI, and Kvasir datasets. (a) Image. (b) Ground Truth. (c) TransUnet. (d) TransFuse. (e) UCTransNet. (f) Ours.

variations in tissue appearance can convey crucial diagnostic information. The remarkable performance of our method can be attributed to the incorporation of attention mechanism, which effectively reduces over-segmentation. It allows the model to focus on relevant regions, mitigating the influence of irrelevant features and noise, thus improving the overall robustness and generalizability.

4.5. Ablation studies

In this section, we assess the impact of the proposed components on segmentation performance across three medical image datasets: ISIC, Kvasir, and BUSI. We conducted experiments using various training settings: fine-tuning with a frozen encoder (Vanilla), fine-tuning with an IPF adapter, fine-tuning with an MDT adapter, and fine-tuning with all adapters. To ensure fair comparison, all these models use SAM pre-trained weights. The experimental results are shown in Table 3. The baseline model, comprising 90.32 million parameters, demonstrates moderate performance on the ISIC dataset with a DICE score of 77.99. However, it underperforms on the Kvasir and BUSI datasets, particularly with DICE scores of 44.74 and 43.11, respectively. The reason for the lower performance is that the SAM image encoder is frozen, and the knowledge it retains is primarily from the natural image domain, resulting in poor generalization and reduced performance in medical image segmentation tasks.

Table 3

Trans-SAM ablation Studies on three medical image datasets.

Methods	Modules	#P(M)	ISIC	Kvasir	BUSI
Vanilla	–	90.32	77.99	44.74	43.11
+IPF	–	+0.470	82.39	72.87	60.35
	+Emb	+0.489	88.15	80.26	68.57
+MDT	–	+0.493	90.61	84.89	71.27
	+SA	+5.675	91.12	86.19	71.77
+MDT	–	+6.152	91.56	88.37	72.07
	+SA	+6.152	91.56	88.37	72.07
Trans-SAM	✓	+6.645	91.89	89.09	73.01

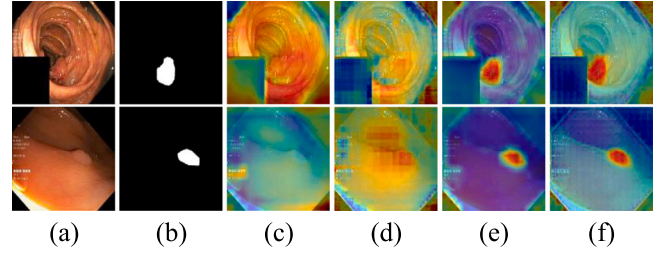


Fig. 6. Visualization results on the Kvasir dataset. (a) Image. (b) Ground Truth. (c) SAM Image Encoder. (d) SAM Mask Decoder. (e) Trans-SAM Image Encoder (f) Trans-SAM Mask Decoder. The red regions indicate the focal points of model, where darker colors signify higher attention concentration. Conversely, deeper blue regions represent dispersed attention, indicating lower focus.

However, after introducing the initial IPF, the newly added fine-tuning parameters provide the image encoder with the feature knowledge required for medical image segmentation, resulting in a significant improvement in performance. Performance improved considerably across three datasets. After adding the initial embedding (Emb) of shared linear output, the model gained supplementary initial features for medical images, leading to performance improvements, with results of 88.15, 80.26, and 68.57 on the three datasets, respectively. Finally, after incorporating the lightweight attention mechanism (TA), performance was further enhanced, reaching 90.61, 84.89, and 71.27 on the three datasets, respectively, demonstrating the effectiveness of the attention mechanism. The second component, “+MDT”, increased the number of parameters by 5.675 M and 6.152 M before and after adding attention, respectively, and also completed rapid convergence during training and improved performance on all datasets.

Furthermore, Combining both components delivering the highest segmentation performance. The DICE score for the ISIC dataset reaches an impressive 91.89. This trend continues across the Kvasir and BUSI datasets, where the Trans-SAM model surpasses all modification in terms of DICE. Overall, the ablation study provides clear evidence that each component contributes positively to the segmentation performance. These enhancements are consistent across various datasets, underscoring the robustness and adaptability of the Trans-SAM for medical image analysis tasks.

In addition, we visualized the attention heat maps after incorporating IPF and MDT adapter. As shown in Fig. 6, we selected two challenging images from the Kvasir dataset, one with a small polyp region and another with low contrast between the polyp region and surrounding tissues. For difficult segmentation cases, the image encoder of the original SAM tends to focus on the background, causing an attention shift in the decoder and ultimately leading to suboptimal segmentation outcomes. However, the image encoder of Trans-SAM can accurately locate the polyp, enabling the mask decoder to refine and enhance the target region, producing accurate segmentation results. These visualizations clearly emphasize the enhanced ability of Trans-SAM to focus on critical features within complex medical imagery, thereby improving the model’s segmentation accuracy in challenging clinical scenarios.

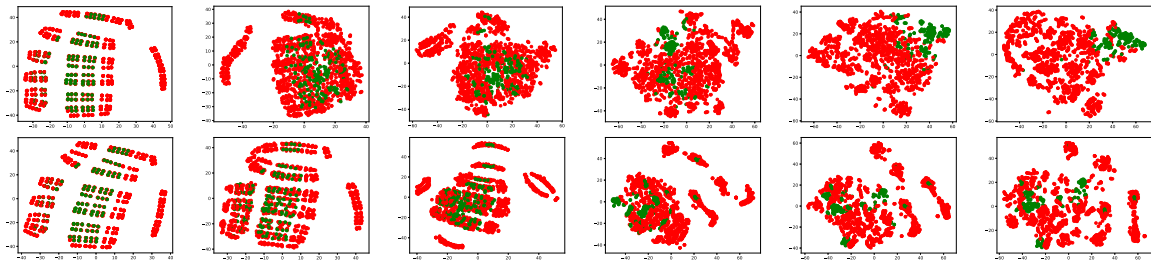


Fig. 7. Visualization of feature distribution using t-SNE on various layers of the image encoder. From left to right, these images correspond to layers 1, 3, 5, 8, 10 and 12; the first row shows the feature distribution post the Trans-SAM image encoder, the second row represents the post SAM image encoder.

Table 4

Performance analysis of SAM and Trans-SAM under different labeled ratios.

Labeled ratio	Methods	ISIC			Kvasir			BUSI		
		DICE (%)	IOU (%)	ACC (%)	DICE (%)	IOU (%)	ACC (%)	DICE (%)	IOU (%)	ACC (%)
5%	SAM	77.55	64.39	83.61	28.16	17.53	75.13	11.56	6.18	53.82
	Trans-SAM	86.25 ↑	77.57 ↑	87.08 ↑	61.65 ↑	45.39 ↑	71.88 ↑	65.26 ↑	50.71 ↑	61.66 ↑
10%	SAM	86.18	76.53	85.75	41.98	25.86	52.68	9.22	4.88	41.87
	Trans-SAM	86.28 ↑	77.69 ↑	87.19 ↑	75.36 ↑	62.19 ↑	78.50 ↑	67.01 ↑	52.13 ↑	62.82 ↑
20%	SAM	83.49	73.15	85.11	75.52	61.38	77.89	15.38	8.88	14.46
	Trans-SAM	86.17 ↑	78.06 ↑	88.13 ↑	78.67 ↑	66.96 ↑	80.81 ↑	70.33 ↑	55.96 ↑	66.69 ↑
30%	SAM	85.92	77.63	88.75	78.18	66.99	78.15	44.05	29.58	36.37
	Trans-SAM	88.33 ↑	80.29 ↑	91.86 ↑	83.08 ↑	71.79 ↑	82.17 ↑	70.39 ↑	56.28 ↑	66.76 ↑
100%	SAM	88.06	78.89	91.23	80.25	68.11	81.57	56.83	43.15	52.78
	Trans-SAM	91.89 ↑	84.35 ↑	92.78 ↑	89.09 ↑	79.15 ↑	89.77 ↑	73.01 ↑	58.85 ↑	69.61 ↑

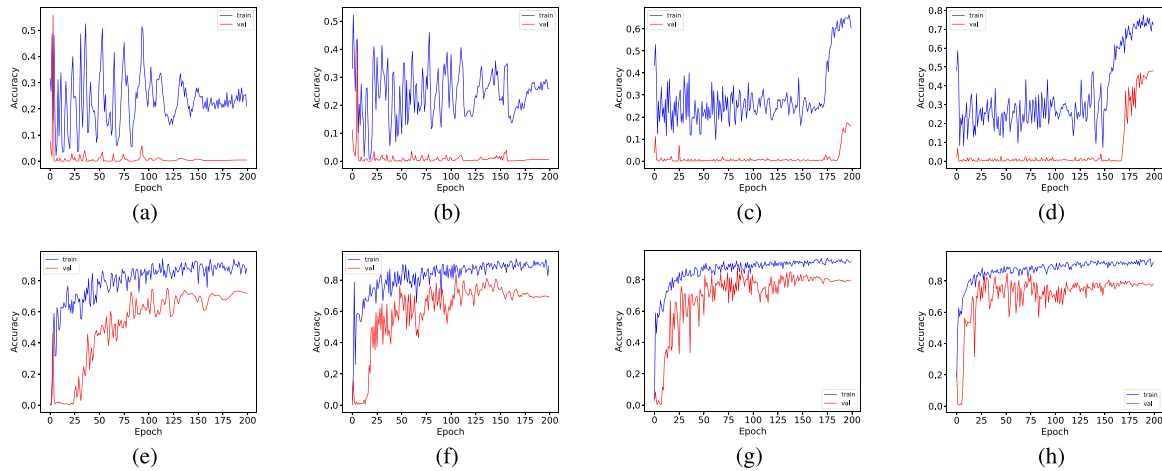


Fig. 8. The change in accuracy of Trans-SAM on the BUSI dataset. From left to right, the curve corresponds to labeled ratios of 5%, 10%, 20%, and 30%. (a)–(d) represent the performance of the original SAM. (e)–(h) represent the performance of Trans-SAM.

4.6. Visualization analysis

In order to more intuitively demonstrate the effect of Trans-SAM, we employed t-SNE [72] to comparatively evaluate the feature extraction and representation capabilities of Trans-SAM and SAM in BUSI segmentation tasks. In Fig. 7, red points represent the target category (e.g., lesion areas), while green points indicate the background category (e.g., normal tissue). From Fig. 7, we can observe distinctive characteristics between the Trans-SAM and SAM regarding feature distribution: (1) **Intra-class Feature Cohesion**: Trans-SAM forms compact clusters of the target category (red points) across different layers, indicating its superior ability to discriminate the target class within the high-dimensional feature space. In contrast, the SAM displays a mixture of the target category with the background, lacking clear aggregation, which results in blurred boundaries between classes. (2) **Inter-class Feature Separation**: At the deeper layers of the encoder, Trans-SAM

visualizations reveal a clear demarcation between the target and background categories, maintaining this separation even where target and background are in close proximity. Conversely, the SAM show significant overlap between the target and background, demonstrating a lack of clear separation. (3) **Consistency and Variability of Feature Expression**: The pattern of separation maintained by Trans-SAM signifies its consistent discriminative power throughout the network layers. On the other hand, the SAM shows substantial variation, which could lead to instability on performance during the training.

4.7. Efficiency analysis

To investigate the data efficiency of Trans-SAM, we evaluated the performance of the original SAM and our method at different labeled ratios.

Table 4 summarizes the results of these evaluations conducted on three medical datasets. It is noteworthy that the Trans-SAM consistently

yielded substantial performance enhancements across all labeled ratios for all datasets. The ISIC dataset, characterized by relatively low color and object shape variations, allowed SAM to exhibit reasonably good performance even when trained with a limited labeled data. Specifically, SAM attained 88% dice of the performance with a fully labeled training set when trained on just 5% labeled data. However, on the Kvasir and BUSI datasets, SAM displayed a noticeable performance decline, particularly when the labeled ratio dropped below 20%. In contrast, the Trans-SAM resulted in a substantial performance boost for SAM. For instance, on the ISIC dataset, Trans-SAM achieved an impressive 94% of the performance level of a fully labeled ratio when trained on only 5% labeled data. Similarly, Trans-SAM showcased a significant advantage on the other two datasets, maintaining outstanding performance even at labeled ratio below 10%. Remarkably, Trans-SAM consistently demonstrated substantial performance improvements across labeled ratios of 5%, 10%, 20%, 30%, and 100%.

In addition, we visualized the curve of Trans-SAM's accuracy with different labeled ratio on BUSI dataset, as shown in Fig. 8. The blue curves represent the changes on training accuracy, while the red curves represent the changes on validation accuracy. From Fig. 8, it can be noted that at labeled ratio below 10%, the original SAM exhibits significant fluctuations on training accuracy. It fails to achieve satisfactory convergence. In contrast, Trans-SAM has brought about significant improvements in the training process. With labeled ratio below 10%, Trans-SAM consistently achieves convergence compared with original SAM. Furthermore, the early stages of training show a smoother and more stable increase in validation accuracy. Moreover, when the labeled ratio exceeding 10%, the model rapidly and steadily enters the convergence phase in the early stages. Compared to the original SAM, the Trans-SAM significantly advances the time required to reach convergence and attains higher levels of accuracy.

Moreover, this also demonstrates that when SAM is directly fine-tuned, knowledge from the new task may overwrite knowledge from the old task, leading to a sharp decline in the model's performance on previous tasks. By fixing the weights of the original SAM model and adding a new feature processing path, we can retain the core capabilities of the original feature extractor during fine-tuning. This allows the model to leverage prior knowledge for feature extraction even with a limited amount of data, achieving good performance and partially alleviating the issue of catastrophic forgetting.

5. Discussion

Currently, Trans-SAM performs excellently in automatic segmentation tasks but has not yet integrated interactive segmentation features. The primary reason for this is the lack of interactive annotation information (such as clicks, boxes, or text) in medical image datasets, as well as the model's design focus on meeting the needs of large-scale automated analysis. In the future, to enhance the model's flexibility and applicability, we will attempt to combine interactive segmentation and automatic segmentation, integrating the two capabilities organically. Specifically, by opening the input part of the prompt encoder, we will allow users to provide prompts through clicking, drawing boxes, or inputting semantic text, which will generate the initial segmentation results. In subsequent steps, the automatic segmentation module will complement the unprompted regions and dynamically update the segmentation masks of the prompted regions to improve overall accuracy. Additionally, a lightweight attention mechanism can be employed to efficiently combine interactive prompts with automatic segmentation, ensuring an optimal balance between response speed and resource consumption. Through an interactive feedback mechanism, users can correct the model's erroneous segmentations, achieving higher efficiency and precision. In the future, we will further explore how to flexibly implement this method in various medical application scenarios to enhance its practical effectiveness in clinical applications.

Despite the strong performance of Trans-SAM across most medical image segmentation tasks, the model faces challenges in tasks where the target regions exhibit significant variability or where the contrast between the target region and surrounding tissues is minimal. For instance, in datasets such as ISIC (skin lesion images), BUSI (breast ultrasound images), and Kvasir (endoscopic images), the target regions are highly variable, and the contrast between the region of interest and surrounding tissues is often very low. In these cases, while Trans-SAM outperforms many existing models, its performance still unsatisfactory in some instances, with the segmentation results for certain images showing considerable confusion in identifying target regions. This issue highlights the challenge of generalizing to medical images with complex, low-contrast, and highly variable regions, which is a direction for future improvement.

Moreover, While Trans-SAM significantly reduces the number of trainable parameters during training through parameter-efficient fine-tuning, the overall parameter size during inference remains relatively large. This can lead to higher memory and computational demands during the inference phase, which may limit the practical deployment of the model in resource-constrained environments. To address this, future work will explore model quantization techniques to further compress the model and improve its efficiency without sacrificing performance.

6. Conclusion

In this paper, we proposed the visual fine tuning strategy to transfer SAM to the medical image segmentation. We have achieved automatic semantic segmentation of medical images using improved SAM and achieved significant accuracy. Trans-SAM investigates the possibility of using convolutions in adapters and demonstrates the effectiveness of applying attention mechanisms within adapters. By employing the proposed visual fine tuning strategy on SAM with a few parameters, we significantly improved performance of SAM in medical image segmentation tasks. We conducted experiments on various medical image segmentation datasets. And the experimental results demonstrated that our proposed method outperforms the majority of existing medical image segmentation models. Additionally, we compared Trans-SAM with other PEFT methods, and the results showed that Trans-SAM also exhibits substantial advantages.

CRedit authorship contribution statement

Yanlin Wu: Writing – original draft, Methodology, Formal analysis. **Zhihong Wang:** Validation, Investigation. **Xiongfeng Yang:** Visualization, Investigation. **Hong Kang:** Formal analysis, Data curation. **Along He:** Supervision, Investigation, Funding acquisition. **Tao Li:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors are unable or have chosen not to specify which data has been used.

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